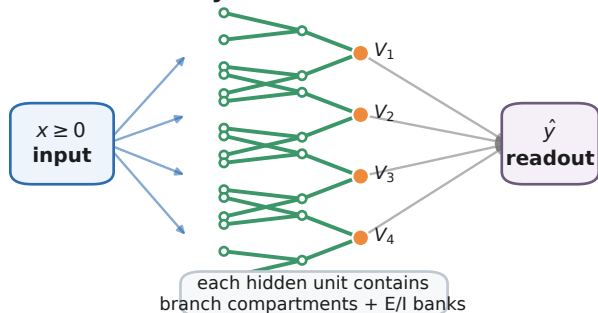
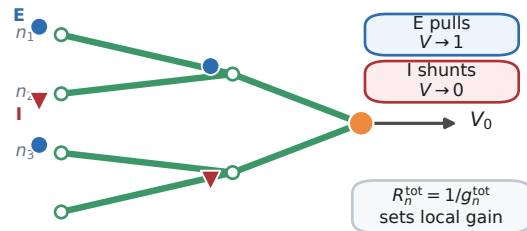


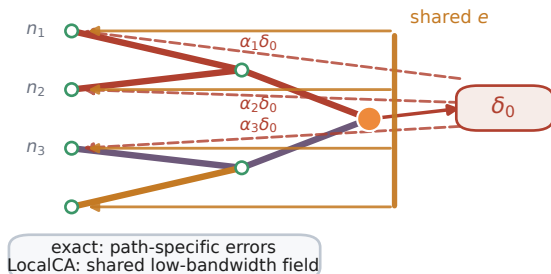
A Network layer: dendritic E units



B Single conductance tree



C Path-specific vs. shared credit



D LocalCA substitutes only error

exact local eligibility

$$\frac{\partial L}{\partial g_i} = x_i R_n^{\text{tot}} (E_i - V_n) \delta_n$$

LocalCA update

$$\Delta g_i \propto x_i R_n^{\text{tot}} (E_i - V_n) e_n$$

feedback object

rank-1 $e_n = \bar{\delta} \mathbf{1}$

neuron $e_n = \delta_0$

rank-K $e_n = Q_n P_K \delta_0$

path $e_n = \Gamma_n(\delta_0)$

oracle $e_n = \tilde{\alpha}_n \delta_0$

same synapse-local eligibility; different non-local feedback bandwidth